

Blade flapping dynamics with fuselage translation and rotation

Recap: Flapping eq. of motion when the helicopter has a body rate q , hover

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} + \Omega^2 \beta = -2q\Omega \sin \psi + \frac{\gamma}{8} q\Omega \cos \psi + \frac{\gamma}{8} \Omega^2 \left[\theta - \frac{4\lambda_i}{3} \right]$$

coriolis effect

aerodyn. effect as a
result of q

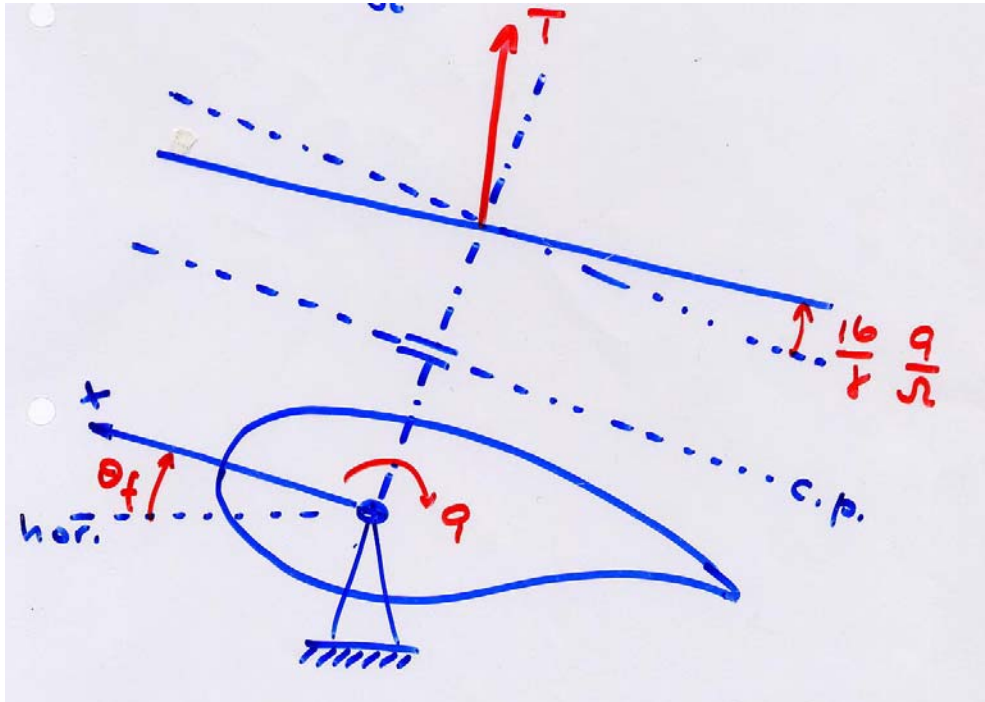
aerodyn. effect as a
result of lift

Transforming to steady-state motion

$$\beta = a_0 - a_1 \cos \psi - b_1 \sin \psi$$



Flapping motion in the non-rotating system when the helicopter has a body rate q – hovering helicopter



$$a_0 = \frac{\gamma}{8} \left[\theta - \frac{4\lambda_i}{3} \right] \quad \text{coning}$$

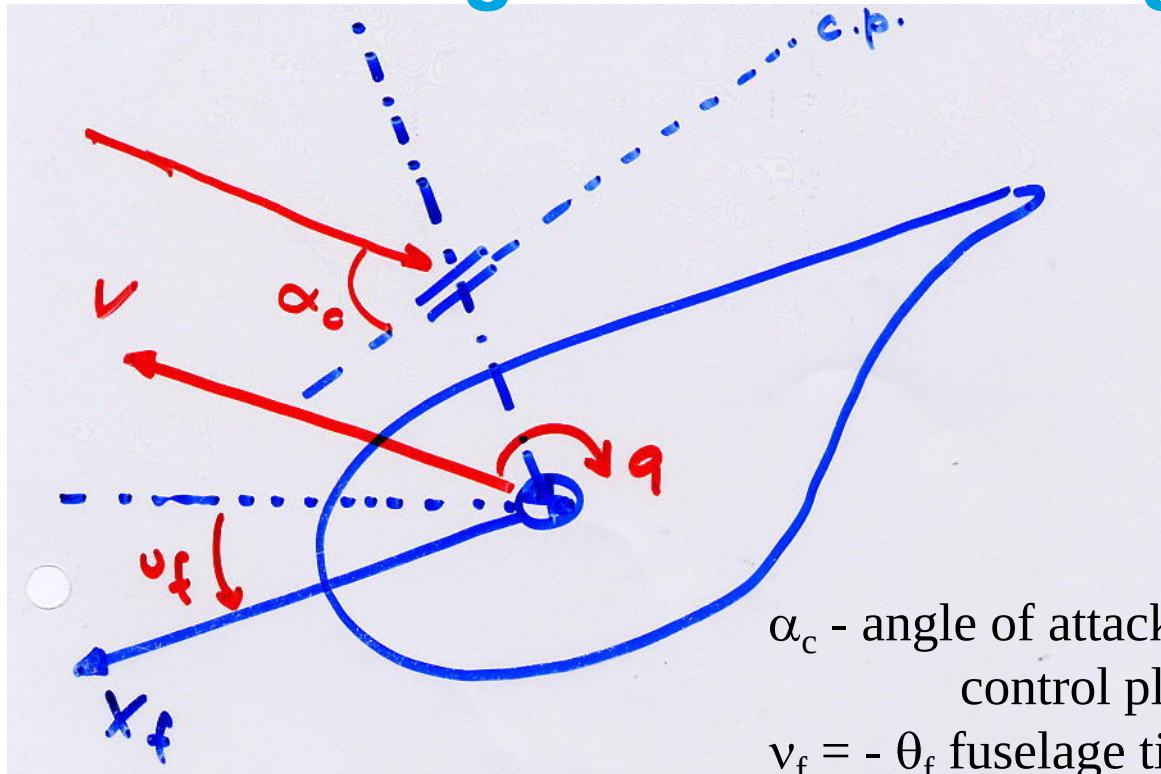
$$a_1 = -\frac{16}{\gamma} \frac{q}{\Omega} \quad \text{longitudinal tilt}$$

$$b_1 = -\frac{q}{\Omega} \quad \text{lateral tilt}$$

Conclusion:

Flapping motion damps the fuselage motion!!!

Flapping motion when the fuselage is rotating and translating (V, q)

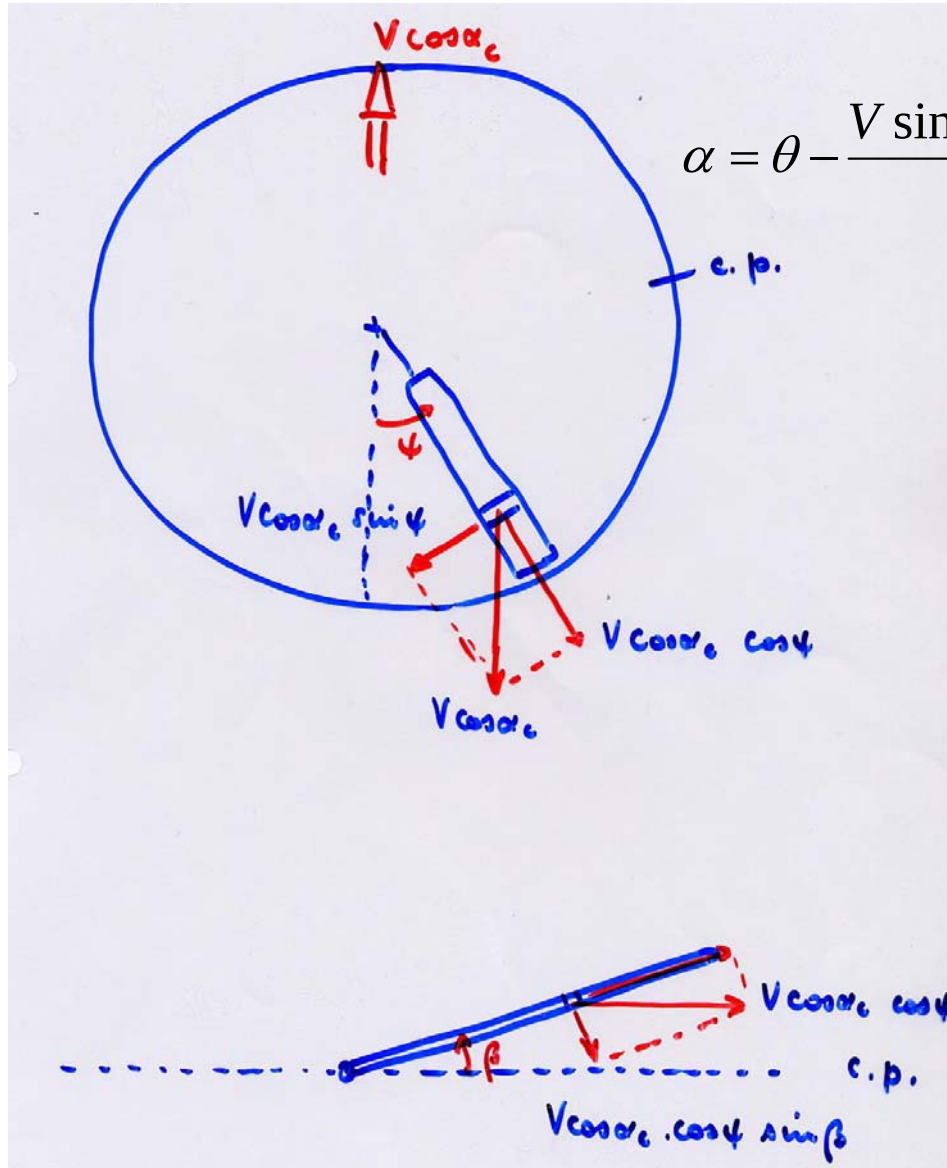


Extra effect of V on a_1

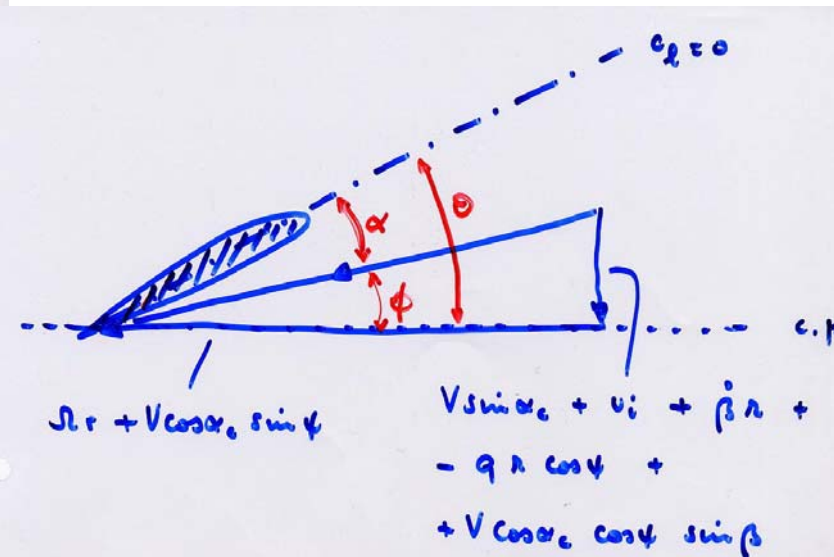
$$\ddot{\beta} + \Omega^2 \beta = -2q\Omega \sin \psi + M_a / I_{bl}$$

$M_a = ?$

Velocities of the blade when the helicopter translates and rotates (V,q)



$$\alpha = \theta - \frac{V \sin \alpha_c + v_i + \dot{\beta} r - q r \cos \psi + V \cos \alpha_c \cos \psi \sin \beta}{\Omega r + V \cos \alpha_c \sin \psi}$$



Interpretation of the aerodynamic flapping moment contribution (helicopter in forward flight)

Excitation
 $\dot{\beta} \Rightarrow \Delta\alpha = -\frac{\dot{\beta}r}{\Omega r + V \cos\alpha_c \sin\psi} \Rightarrow \Delta L = C_{l\alpha}\Delta\alpha \frac{1}{2}\rho(\Omega r)^2 c dr \Rightarrow \Delta M_a = \int_0^R \Delta L r$

Aerodynamic forces have the character of a damping term in the flapping equation. Compared to hover, in forward flight this term is time dependent through ψ

However, averaged over one revolution the mean damping is the same as in the hovering case

In forward flight the dominating excitation of the blade flapping is the asymmetry of the flow encountered by a blade

Aerodynamic flapping moment for the helicopter in forward flight

$$M_a = \int_0^R C_{l\alpha} \alpha \frac{1}{2} \rho (\Omega r + V \cos \alpha_c \sin \psi)^2 c dr \cdot r$$

$$\alpha = \theta - \frac{V \sin \alpha_c + v_i + \dot{\beta} r - q r \cos \psi + \beta V \cos \alpha_c \cos \psi}{\Omega r + V \cos \alpha_c \sin \psi}$$

Non-dimensional
notations

$$\mu = \frac{V \cos \alpha_c}{\Omega R} \quad \lambda_c = \frac{V \sin \alpha_c}{\Omega R} \quad \lambda_i = \frac{v_i}{\Omega R}$$

$$M_a / I_{bl} = \frac{\gamma}{8} \Omega^2 \left[\theta(1 + \mu^2) - \frac{4}{3}(\lambda_c + \lambda_i) + \sin \psi(\dots) + \cos \psi(\dots) + \sin 2\psi(\dots) + \cos 2\psi(\dots) \right. \\ \left. - \beta \mu \left(\frac{4}{3} \cos \psi + \mu \sin 2\psi \right) - \frac{\dot{\beta}}{\Omega} \left(1 + \frac{4}{3} \mu \sin \psi \right) \right]$$

Flapping equation of motion when the helicopter translates and rotates (V,q)

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} \left[1 + \frac{4}{3} \mu \sin \psi \right] + \Omega^2 \beta \left[1 + \frac{\gamma}{6} \mu \cos \psi + \frac{\gamma}{8} \mu^2 \sin 2\psi \right] =$$

Aero damping **Centrifugal spring + Aero spring**

$$-2q\Omega \sin \psi + \frac{\gamma}{8} \Omega^2 \theta (1 + \mu^2) - \frac{\gamma}{6} \Omega^2 (\lambda_c + \lambda_i) + \frac{\gamma}{8} \Omega^2 \mu \sin \psi \left(\frac{8}{3} \theta - 2(\lambda_c + \lambda_i) \right)$$

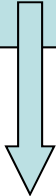
$$+ \frac{\gamma}{8} \Omega^2 \frac{q}{\Omega} \cos \psi + \frac{\gamma}{12} \Omega^2 \frac{q}{\Omega} \mu \sin 2\psi - \frac{\gamma}{8} \Omega^2 \theta \mu^2 \cos 2\psi$$

Excitation function of $(\theta, \lambda_c, \lambda_i, \mu, q)$

Conclusion on the flapping motion in forward flight

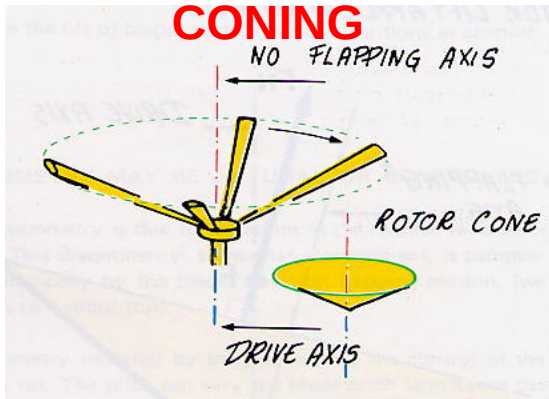
- 2nd order linear differential equation with variable coeffs.
- excitation in 1-P and 2-P
- periodic equation with periodic frequency => Floquet theory
- even with positive damping in the system, there is a possibility for instability
- frequency of excitation $\sim$ natural frequency

Transformation to the nonrotating system

$$\beta = a_0 - a_1 \cos \psi - b_1 \sin \psi$$


Substitute this in the flapping equation in rotating system from slide 8

Disc tilt motion Fuselage translation and rotation

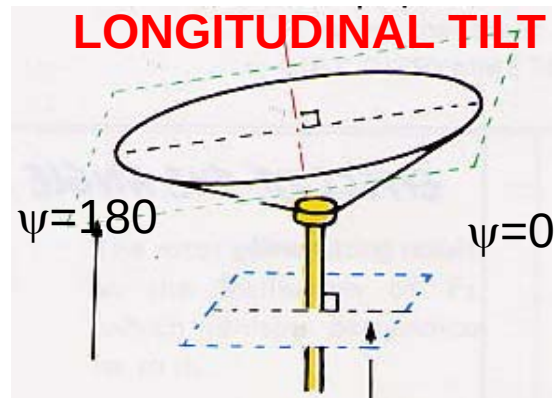


Coning angle:

$$a_0 = \frac{\gamma}{8} \left[\theta(1 + \mu^2) - \frac{4}{3}(\lambda_i + \lambda_c) \right]$$

Longitudinal disc tilt:

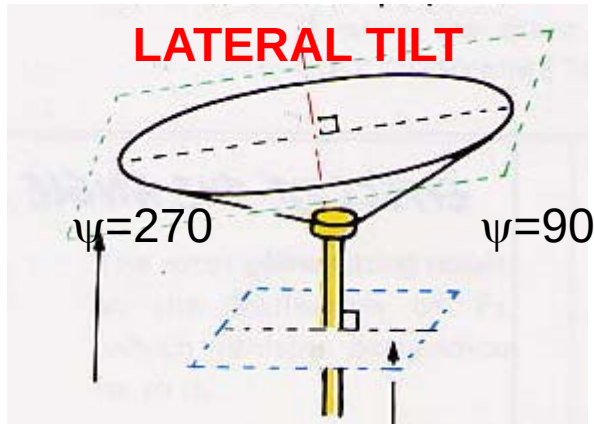
The rotor cone is blowing backward



$$a_1 = \frac{-\frac{16}{\gamma} \frac{q}{\Omega} + \frac{8}{3} \mu \theta - 2\mu(\lambda_c + \lambda_i)}{1 - 1/2 \mu^2} \quad fct(\mu, \lambda_c, \lambda_i, \theta, q / \Omega)$$

$a_1 > 0$ cone tilting back

Disc tilt motion Fuselage translation and rotation



Lateral disc tilt

The rotor cone is tilting to the right

$$b_1 = \frac{-\frac{q}{\Omega} + \frac{4}{3}\mu a_0}{1 + 1/2\mu^2}$$

$b_1 > 0$ cone tilting to the right

In reality lateral tilt is strongly influenced by the non-uniformity of the induced velocity $\Rightarrow$ an extra term K' is added to lateral disc tilt

$$b_1 = \frac{-\frac{q}{\Omega} + \frac{4}{3}\mu a_0}{1 + 1/2\mu^2} + \frac{K' \lambda_i}{1 + 1/2\mu^2} \quad K' = \frac{1.33 \left| \mu / (\lambda_c + \lambda_i) \right|}{1.2 + \left| \mu / (\lambda_c + \lambda_i) \right|}$$

Summing up: Flapping motion in rotating system

$$\ddot{\beta} + \Omega^2 \beta = M_a / I$$

1. No Fuselage motion (only aerodynamic excitation)

- Hovering:

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} + \Omega^2 \beta = \frac{\gamma}{8} \Omega^2 \left[\theta - \frac{4\lambda_i}{3} \right] \xrightarrow{\text{T.P.P.}} a_0$$

- Hovering + vertical gust:

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} + \Omega^2 \beta = \frac{\gamma}{8} \Omega^2 \left[\theta - \frac{4\lambda_i}{3} \right] + \frac{|M_a|}{I} \cos \Omega t$$

2. Fuselage rotating with pitch rate q

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} + \Omega^2 \beta = -2q\Omega \sin \Omega t + \frac{\gamma}{8} q\Omega \cos \Omega t + \frac{\gamma}{8} \Omega^2 \left[\theta - \frac{4\lambda_i}{3} \right]$$

coriolis effect

aerodyn. effect as a
result of q

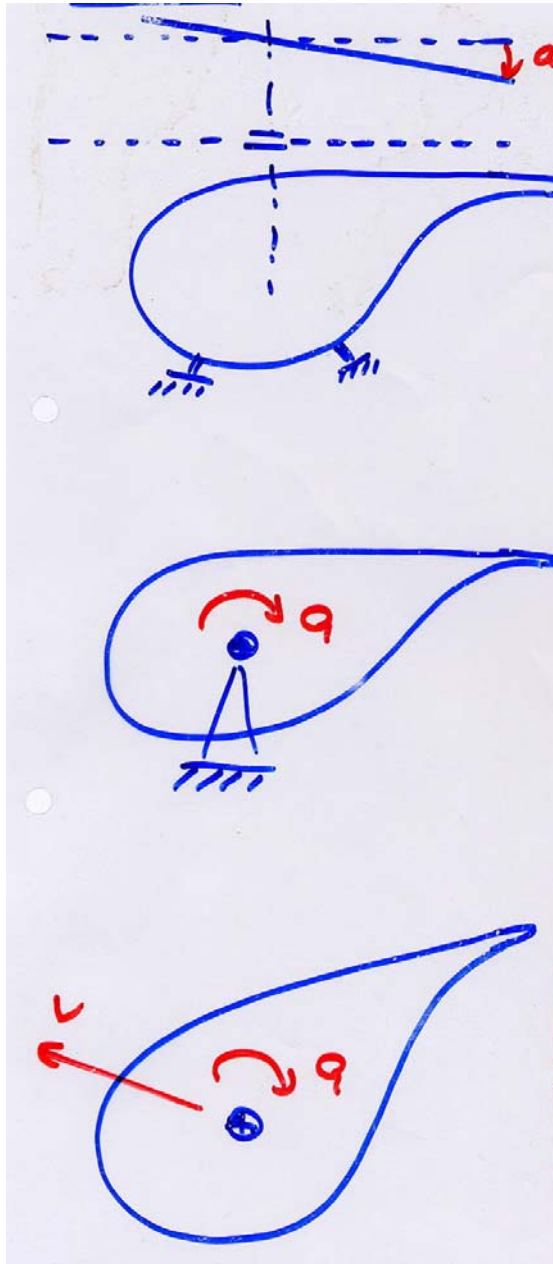
aerodyn. effect as a
result of lift

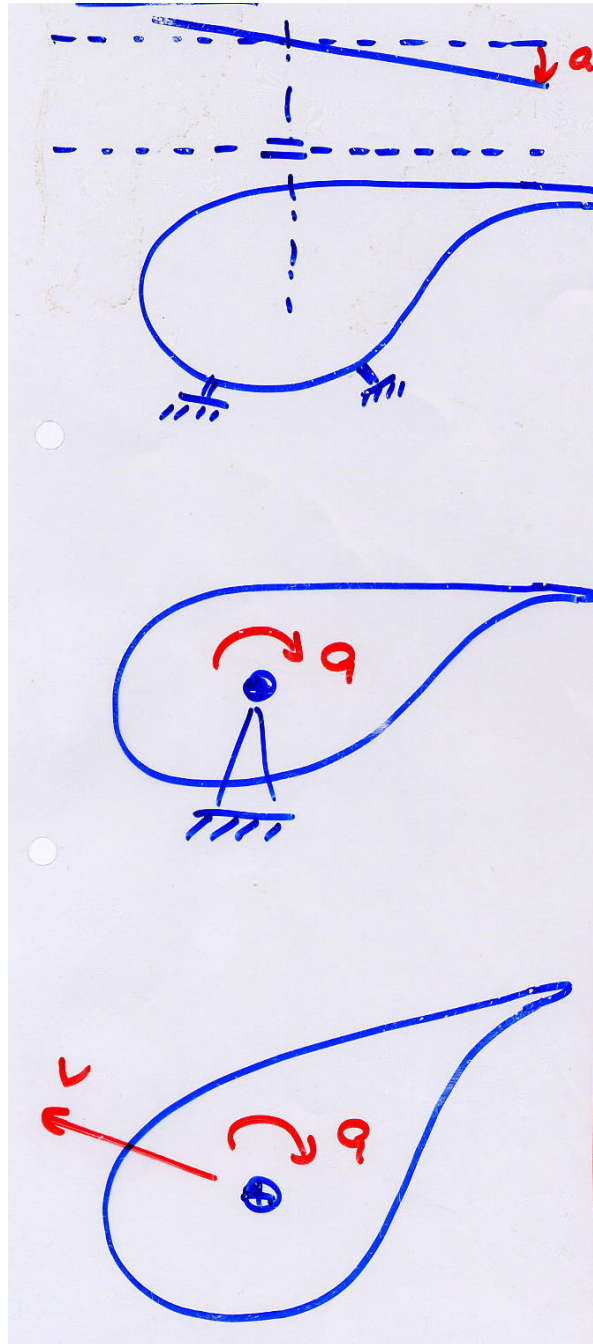
T.P.P.
 a_0, a_1, b_0

3. Fuselage rotating and translating (V, q)

$$\ddot{\beta} + \frac{\gamma}{8} \Omega \dot{\beta} \left[1 + \frac{4}{3} \mu \sin \psi \right] + \Omega^2 \beta \left[1 + \frac{\gamma}{6} \mu \cos \psi + \frac{\gamma}{8} \mu^2 \sin 2\psi \right] =$$

$$\{ \dots \} + \sin \psi \{ \dots \} + \cos \psi \{ \dots \} + \sin 2\psi \{ \dots \} + \cos 2\psi \{ \dots \}$$





Summing up: Flapping motion in non-rotating system- a1

1. No Fuselage motion

$$a_1 = 0 \quad a_0 = \frac{\gamma}{8} \left[\theta - \frac{4\lambda_i}{3} \right] \quad b_1 = 0$$

2. Fuselage rotation

$$a_1 = -\frac{16}{\gamma} \frac{q}{\Omega} \quad \text{Backward tilt}$$

$$a_0 = \frac{\gamma}{8} \left[\theta - \frac{4\lambda_i}{3} \right] \quad b_1 = -\frac{q}{\Omega}$$

Flapping is damping the fuselage motion (is stabilizing the motion)

3. Fuselage rotation and translation

$$a_1 = \frac{-\frac{16}{\gamma} \frac{q}{\Omega} + \frac{8}{3} \mu \theta - 2\mu(\lambda_c + \lambda_i)}{1 - 1/2\mu^2}$$

Flapping is increasing the nose-up fuselage pitch (is destabilizing the motion)

Backward tilt
(Backward blowing)